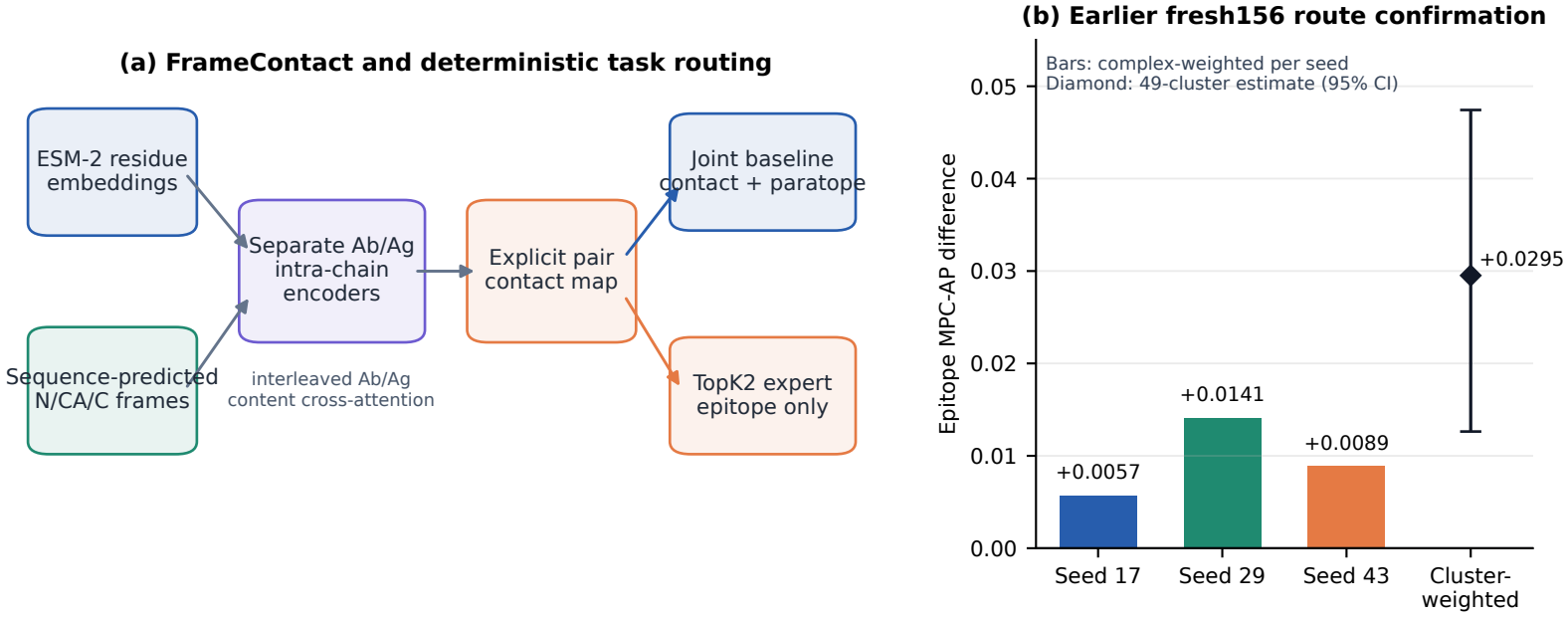
(a) FrameContact and deterministic task routing
ESM-2 residue embeddings
Sequence-predicted N/CA/C frames
Separate Ab/Ag intra-chain encoders
interleaved Ab/Ag content cross-attention
Explicit pair contact map
Joint baseline contact + paratope
TopK2 expert epitope only
(b) Earlier fresh156 route confirmation
Bars: complex-weighted per seed
Diamond: 49-cluster estimate (95% CI)
Epitope MPC-AP difference
0.00
0.01
0.02
0.03
0.04
0.05
+0.0057
+0.0141
+0.0089
+0.0295
Seed 17
Seed 29
Seed 43
Cluster-weighted